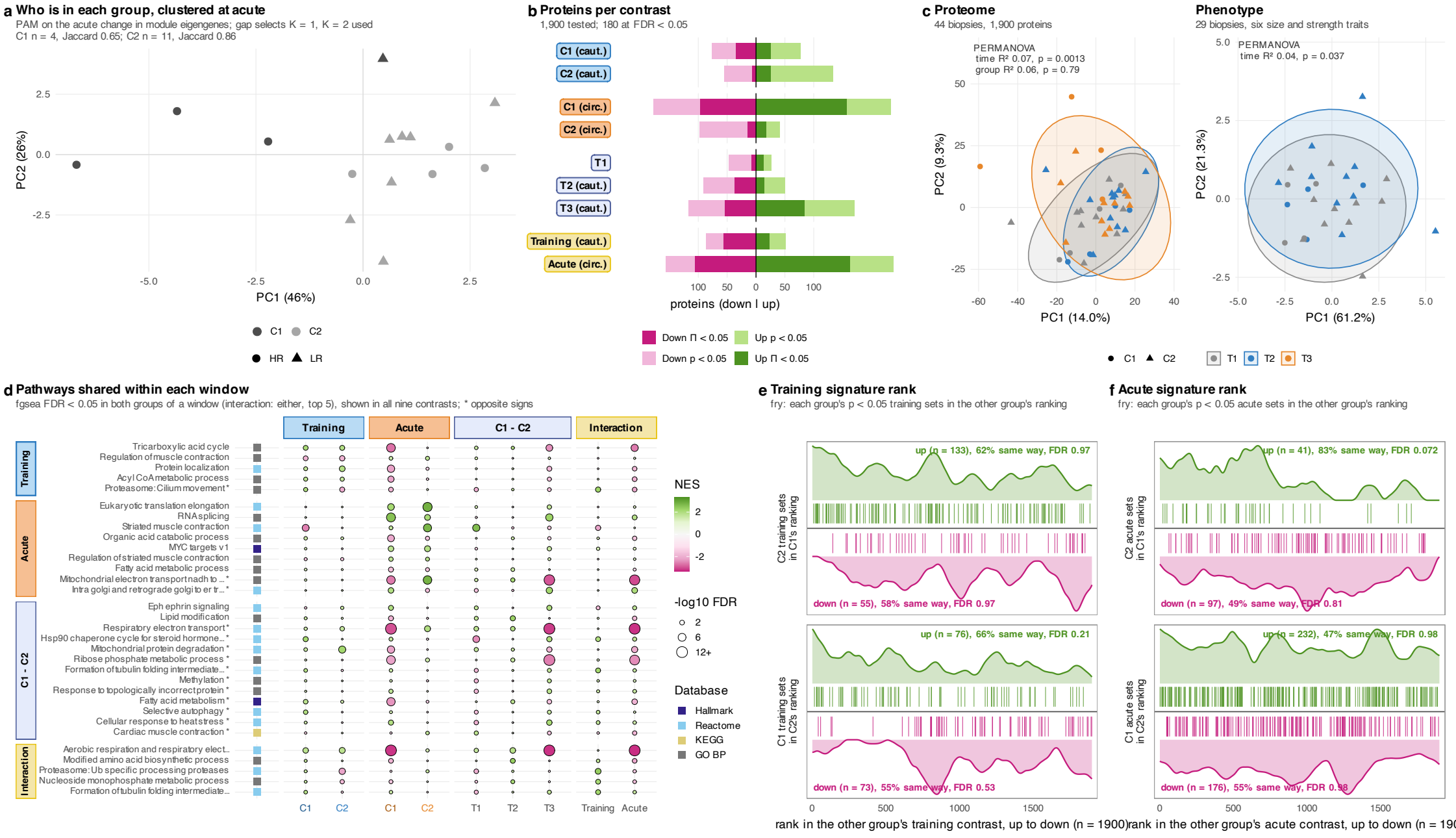

a Who is in each group, clustered at acute
PAM on the acute change in module eigengenes; gap selects K = 1, K = 2 used
C1 n = 4, Jaccard 0.65; C2 n = 11, Jaccard 0.86
PC2 (26%)
PC1 (46%)
C1
C2
HR
LR
b Proteins per contrast
1,900 tested; 180 at FDR < 0.05
C1 (caut.)
C2 (caut.)
C1 (circ.)
C2 (circ.)
T1
T2 (caut.)
T3 (caut.)
Training (caut.)
Acute (circ.)
proteins (down | up)
Down Π < 0.05
Up p < 0.05
Down p < 0.05
Up Π < 0.05
c Proteome
44 biopsies, 1,900 proteins
PERMANOVA
time R² 0.07, p = 0.0013
group R² 0.06, p = 0.79
PC2 (9.3%)
PC1 (14.0%)
Phenotype
29 biopsies, six size and strength traits
PERMANOVA
time R² 0.04, p = 0.037
PC2 (21.3%)
PC1 (61.2%)
C1
C2
T1
T2
T3
d Pathways shared within each window
fgsea FDR < 0.05 in both groups of a window (interaction: either, top 5), shown in all nine contrasts; * opposite signs
Training
Acute
C1 - C2
Interaction
Tricarboxylic acid cycle
Regulation of muscle contraction
Protein localization
Acyl CoA metabolic process
Proteasome: Cilium movement*
Eukaryotic translation elongation
RNA splicing
Striated muscle contraction
Organic acid catabolic process
MYC targets v1
Regulation of striated muscle contraction
Fatty acid metabolic process
Mitochondrial electron transport nadh to ... *
Intra golgi and retrograde golgi to er tr... *
Eph ephrin signaling
Lipid modification
Respiratory electron transport*
Hsp90 chaperone cycle for steroid hormone... *
Mitochondrial protein degradation *
Ribose phosphate metabolic process *
Formation of tubulin folding intermediate... *
Methylation *
Response to topologically incorrect protein *
Fatty acid metabolism *
Selective autophagy *
Cellular response to heat stress *
Cardiac muscle contraction *
Aerobic respiration and respiratory elect...
Modified amino acid biosynthetic process
Proteasome: Ub specific processing proteases
Nucleoside monophosphate metabolic process
Formation of tubulin folding intermediate...
C1
C2
C1
C2
T1
T2
T3
Training
Acute
NES
2
0
-2
-log10 FDR
2
6
12+
Database
Hallmark
Reactome
KEGG
GO BP
e Training signature rank
fry: each group's p < 0.05 training sets in the other group's ranking
up (n = 133), 62% same way, FDR 0.97
down (n = 55), 58% same way, FDR 0.97
C2 training sets in C1's ranking
up (n = 76), 66% same way, FDR 0.21
down (n = 73), 55% same way, FDR 0.53
C1 training sets in C2s ranking
0
500
1000
1500
rank in the other group's training contrast, up to down (n = 1900)
f Acute signature rank
fry: each group's p < 0.05 acute sets in the other group's ranking
up (n = 41), 83% same way, FDR 0.072
down (n = 97), 49% same way, FDR 0.81
C2 acute sets in C1's ranking
up (n = 232), 47% same way, FDR 0.98
down (n = 176), 55% same way, FDR 0.98
C1 acute sets in C2s ranking
0
500
1000
1500
rank in the other group's acute contrast, up to down (n = 190